

CH4_Inventory_Main.R

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CH4_inventory_build	<i>Main function to calculate gridded methane emissions. Runs sector by sector according to the config file.</i>
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Description

‘CH4_inventory_build’ runs multiple other functions to calculate the gridded methane emissions sector-by-sector using information provided in the config file and inputs.

Usage

```
CH4_inventory_build(  
  input_directory =  
    "G:/My Drive/Shepson Group Drive/Kris/Philly Inventory/Raw_data_rewrite/",  
  output_directory =  
    "G:/My Drive/Shepson Group Drive/Kris/Philly Inventory/Processed_rewrite/",  
  code_directory = "~/../../Kristian/Desktop/methane_inventory/src/",  
  plot_directory =  
    "G:/My Drive/Shepson Group Drive/Kris/Philly Inventory/Figures_rewrite/",  
  focus_city = "Philadelphia, PA--NJ--DE--MD",  
  inventory_year = 2019,  
  domain = as.data.frame(cbind(c(-76.65, -73.65), c(38.97, 40.97))),  
  domain_res = 0.01,  
  domain_crs = "epsg:4326",  
  ACES_directory =  
    "G:/My Drive/Shepson Group Drive/General Inventories and Shapefiles/Inventories/ACES_v2.0",  
  vulcan_directory =  
    "G:/My Drive/Shepson Group Drive/General Inventories and Shapefiles/Inventories/Vulcan_v3.0",  
  verbose = TRUE  
)
```

Arguments

input_directory	Character providing the full filepath to load/save input data
code_directory	Character providing the full filepath to source all other functions and the config.
plot_directory	Character providing the full filepath to save figures. Only relevant if verbose = TRUE.
focus_city	Character or numeric or a vector of characters or numerics. Each character must exactly match the Census name, which can be found at https://www.census.gov/programs-surveys/geography/guidance/geo-areas/urban-rural.html for the 2020 census and at https://www2.census.gov/geo/pdfs/maps-data/maps/reference/2010UAUC_List.pdf for the 2010 census. Years rely on the naming scheme of the most recent census, so e.g., 2018 would rely on the 2010 names. Each numeric must exactly match the code given to the urban area, also detailed in the above documents. This will be used to subset the urban area census outlines and used in any visualizations.
inventory_year	Numeric indicating the desired year of data to use.
domain	SpatRaster providing the desired output grid, including the desired resolution and coordinate reference system or a 2x2 data frame. For the data frame, the first column would be x values, the second column y values, and the first row would be the minima, the second row the maxima. This is used to build a SpatRaster.
domain_res	Numeric providing the resolution for the domain. Can be length 1 or 2 (x,y).
domain_crs	Character providing the projection of the domain
ACES_directory	Character providing the full filepath to a folder with sectoral Anthropogenic Carbon Emission System (ACES) inventory files that have been averaged into annual files. ACES is available at https://doi.org/10.3334/ORNLDAAAC/1943 .
vulcan_directory	Character providing the full filepath to a folder with annual, sectoral Vulcan inventory files. Vulcan v3.0 is available at https://doi.org/10.3334/ORNLDAAAC/1741 .
verbose	Logical indicating whether to save additional output. This includes easier to read csv files of the data that has been gridded for multiple sectors as well as sector and subsector visuals.

Details

This function will source all other functions and the config, create the domain from the input data in the case that a SpatRaster was not provided, and download some files (Census tigerlines) that will be needed for multiple sectors. It will then use the information in the config file and inputs to run every relevant sector. As such, the inputs to this function and the config may require user editing.

See references [Vulcan](#) and [ACES](#)

Value

Nothing is returned from the function, but there will be frequent user updates as multiple other functions are run and output files created for the various sectors.

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Examples

```
CH4_inventory_build(input_directory="G:/My Drive/Shepson Group Drive/Kris/Philly Inventory/Raw_data_rewrite/",
                    output_directory="G:/My Drive/Shepson Group Drive/Kris/Philly Inventory/Processed_rewrite/",
                    code_directory="~/../Kristian/Desktop/methane_inventory/src/",
                    plot_directory="G:/My Drive/Shepson Group Drive/Kris/Philly Inventory/Figures_rewrite/",
                    focus_city="Philadelphia, PA--NJ--DE--MD",
                    inventory_year=2019,
                    domain=as.data.frame(cbind(c(-76.65,-73.65),
                                                c(38.97,40.97))),
                    domain_res=0.01,
                    domain_crs="epsg:4326",
                    ACES_directory="G:/My Drive/Shepson Group Drive/General Inventories and Shapefiles/Inventories/
vulcan_directory="G:/My Drive/Shepson Group Drive/General Inventories and Shapefiles/Inventories/
                    verbose=TRUE)
```

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